Fuel Economy Report

1974 *Motor Trend* road tests · September 2026 · prepared with tweave

20.09 mpg
AVERAGE FUEL ECONOMY

−0.87
CORRELATION, MPG VS. WEIGHT

33.9 mpg
BEST IN TEST: TOYOTA COROLLA

Weight is the story

Across the 32 cars compiled for *Motor Trend* by Henderson & Velleman (1981), fuel economy falls sharply with vehicle weight. A simple linear model

$$\text{mpg}_i = \beta_0 + \beta_1 \text{weight}_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \text{Normal}(0, \sigma^2) \tag{1}$$

estimates that each additional 1,000 lbs costs about **5.34 mpg** ($R^2 = 0.75$).

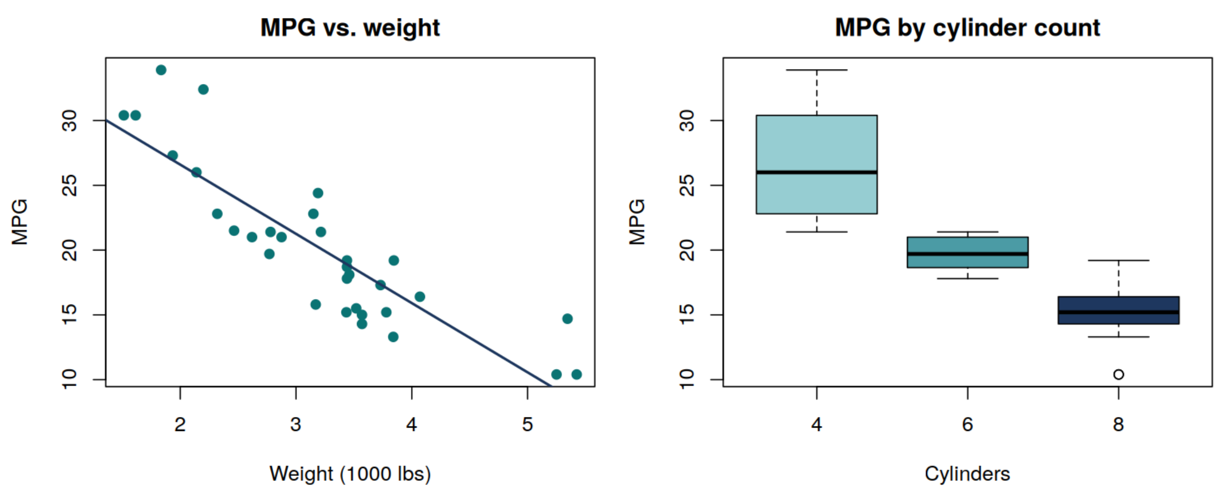

Figure 1: Fuel economy against weight (left, with fitted line) and by cylinder count (right). Heavier, higher-cylinder cars are thirstier.

As Figure 1 shows, the relationship is strong and close to linear over the observed range.

> **Reading note:** the cylinder-count boxplot mostly restates the weight effect — heavier cars have more cylinders. Treat these as one finding, not two.

Leaderboard

The efficiency podium is a clean sweep for light four-cylinder imports — every car in the top five weighs under 2,300 lbs.

Model	MPG	Cylinders	Weight
Toyota Corolla	33.9	4	1.83
Fiat 128	32.4	4	2.20
Honda Civic	30.4	4	1.61
Lotus Europa	30.4	4	1.51
Fiat X1-9	27.3	4	1.94

Does the transmission matter?

+7.24 mpg

Manual-transmission cars average **7.24 mpg more** than automatics in this sample (Welch t-test: $t = 3.77$, $p = 0.0014$).

Before crowning the stick shift, note the confound: the manual cars here are also much lighter[1] – transmission choice and weight travel together in 1974's lineup.

Transmission	Mean MPG	Mean weight
Automatic	17.1	3.77
Manual	24.4	2.41

Discussion

The efficiency leaders are all light four-cylinder imports; the sample mean of $\overline{x} = 20.09$ mpg is dragged down by a tail of heavy V8s, several of which manage barely half that figure.

For 1974 buyers the advice writes itself: weight, not engineering exotica, predicts what you'll spend at the pump. For modern readers, the dataset remains a friendly reminder that a single well-chosen covariate often explains most of the variance. All computation was done in R (R Core Team, 2024).

Bibliography

Henderson, H. V., & Velleman, P. F. (1981). Building multiple regression models interactively. *Biometrics*, *37*(2), 391–411.

R Core Team. (2024). *R: A Language and Environment for Statistical Computing*. https://www.r-project.org/

[1] Mean weight 2.41 vs. 3.77 thousand lbs.